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SULITJELMA GABBRO, NORTHERN NORWAY: A PALAEOMAGNETIC RESULT

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The Sulitjelma Gabbro situated at 67.2°N, 15.4°E was intruded close to the Late Ordovician climax of regional metamorphism in the northern Scandinavian Caledonides. Magnetic properties have been examined from samples collected at seven localities in the south western part of this body. Total NRM directions show a tendency to be aligned near the present earth's magnetic field direction in this region. Stability to a.f. demagnetisation is low and commensurate with low Koenigsberger ratios (0.001–0.16) and the presence of unoxidised magnetite as the principal remanence carrier. After cleaning the site mean directions no longer show an alignment near the present earth's field and of six statistically significant sites three are approximately reversed with respect to remainder. The combined mean direction after cleaning is $D = 195°$, $I = 15°$ (precision parameter $k = 6$) and the derived virtual geomagnetic pole is at 0°E, 14°S ($\alpha_{95} = 23°$). This pole is close to poles of comparable age from the British Isles and suggests that Britain and Norway were part of the same crustal plate in Ordovician times. Discrepancies between Siluro-Devonian results from the two regions may be due to inadequate age coverage of present results.

1. Introduction

The suggestion from geological evidence that the Caledonian fold belt may have resulted from plate collision as a consequence of the closing of a "Proto Atlantic" [1, 2] has prompted a quantitative analysis using palaeomagnetic studies. Within the Euramerican plate the apparent polar wander curves for the North American, Baltic/Russian and British regions are distinct [3] and one possible interpretation is that the first and second regions joined on the site of the Scandinavian Caledonides in Silurian times, while the second and third regions joined on a suture in the North Sea region in Devonian times. Unfortunately Palaeozoic data from Norway prior to the Silurian consist of a single Cambrian pole [4] and we cannot yet be certain whether the Baltic/Russian region formed a single plate or whether the differences [5] present in Siluro-Devonian data from Britain and Norway are long term, and to be interpreted in terms of relative plate movements between the regions.

This note reports a palaeomagnetic result from the Late Ordovician Sulitjelma Gabbro of northern Norway which has some bearing on these problems.

2. Geological setting, sampling and petrology

The Sulitjelma Gabbro outcrops on the Norwegian-Swedish border where it intrudes metasediments and metavolcanics of the Norwegian Caledonides. Early geological knowledge of the region was summarized by Vogt [6] who regarded the gabbro as a syntectonic body intruded contemporaneously with the regional metamorphism. Kautsky [7] considered the rocks of this region to belong to three allochthonous nappes. More recent accounts of the structure of this region are given by Nicholson and Rutland [8] and Mason [9]; the latter author provided a detailed account [10] of the chemistry and structure of the gabbro complex.

On a regional scale the gabbro is broadly concor-

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dant with the schistosity and shows no chilled contact facies; the intrusion postdates the formation of the Gusak nappe and at least one episode of major folding. It is closely spatially-related to the Furuland Granite, whose intrusion it slightly preceded, and was folded into a synclinal structure by an episode which also affected the granite.

The maximum possible age of the Sulitjelma Gabbro from stratigraphic evidence is mid-Ordovician since it intrudes volcanics of this age. Wilson [11] has determined an Rb/Sr age of 443 ± 16 m.y. for the Furuland Granite which is currently the best estimate for the minimum age of the gabbro.

Samples for this study were collected at seven localities from the south western part of the complex in the "flasser gabbro" (a tectonic mélange of gabbro and amphibolite [9]), and the margin of the metagabbro units of Mason [10] between 16°10′E, 67°9′N and 16°15′E, 67°10′N. The gabbro here has undergone both metamorphism to amphibolite facies and considerable deformation along discrete shear surfaces. Lumps were oriented to magnetic co-ordinates while still attached to the outcrop. Weather conditions did not permit the taking of sun sights, but no declination anomalies were noticed, and magnetic intensities of the sampled material suggest that they are unlikely to have been significant. Several cores were extracted from each of 16 such lumps for magnetic studies and representative thin and polished sections were prepared for petrological examination.

All the rocks used in this study are amphibolitised. Green hornblende is the predominant mineral and shows the development of a weak to moderate foliation; feldspars show all stages of alteration from incipient sericitisation to large-scale replacement by clinozoisite. The content of opaque minerals is generally small and in some highly foliated samples (1–3) occur grouped along the foliation. As a rule samples 1–12 show a higher degree of alteration than the remainder and opaque grains are overgrown by hornblende or completely altered to secondary aggregates. Magnetite is present mostly in the size range 10–50 μ with the sulphides pyrite and pyrrhotite also prominent and typically up to an order larger in size. Internal granulation, [12] with marginal and cleavage plane development of maghemite, occurs to some degree in most samples.

3. Palaeomagnetic results

Intensities of magnetisation vary between 0.1×10^{-3} and 0.4×10^{-6} gauss with a log mean of 0.1×10^{-4} gauss; samples from the "flaser gabbro" generally have intensities about an order lower than those from the metagabbro. The bulk of the total NRM's group with steep positive inclinations and variable declinations (Fig. 1d) suggesting that they possess components acquired from the present earth's field. One sample from each lump was treated by a.f. demagnetisation in steps of 100 oersteds up to a peak value of 1000 oersteds. Selected demagnetisation characteristics are shown in Fig. 1a together with the behaviour of the stability index defined by Briden [13]. Samples typically show a progressive fall in moment for the first few hundred oersteds of treatment, followed by erratic behaviour. Acquisition of secondary moments is sometimes evident in directional behaviour (Figs. 1b and 1c). Samples show only low stabilities, the only general tendency being for their directions to move away from the present earth's field. The common correlation noted by several workers [14–16] between titanomagnetite high-temperature oxidation and stability of natural remanence suggests that completely unexsolved magnetite is the predominant remanence carrier in these rocks and this is confirmed by petrological observations. Koenigsberger rations are uniformly low (0.001–0.16, averaging 0.02), commensurate with unstable NRM behaviour.

Most pilot samples show two ranges of high stability indices (Fig. 1a) and the remaining sample from each lump was demagnetised in these fields. A group mean direction for each locality is calculated by combining site directions which have the best within-sample grouping (Table 1). These localities are: A (samples 19, 20, 22), B (sample 21), C (samples 15 and 16), D (samples 13 and 14), E (samples 9 and 11), F(samples 4, 6 and 7) and G (samples 1–3). The cleaned directions show northerly negative and southerly positive directions which are no longer related to the present earth's field direction. Samples from localities B, D and F are approximately reversed with respect to those from localities A, E and G. Site C is random after cleaning. Reversing sites B, D and F and combining with the remainder yields a mean direction

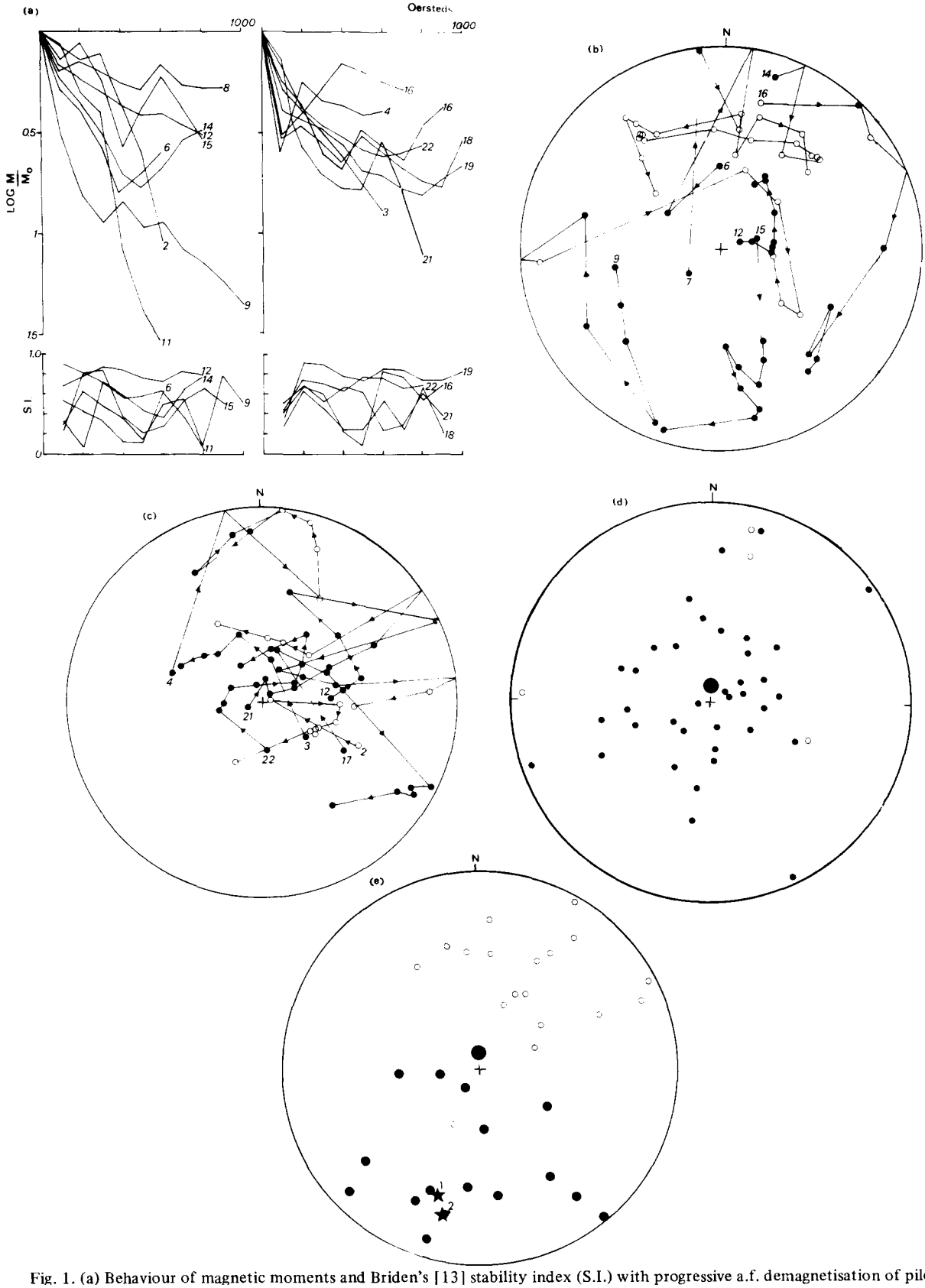


Fig. 1. (a) Behaviour of magnetic moments and Briden's [13] stability index (S.I.) with progressive a.f. demagnetisation of pilot

TABLE 1
Sulitjelma Gabbro, northern Norway (15.4°E, 67.2°N)

Site	Number of samples N.	Total NRM results D	I	R	k	Cleaning fields (oersteds)	Number of samples N.	Results after cleaning D	I	R	k	Palaeomagnetic pole	
A	10	209	44	8.0	5	200–500	10	192	47	7.5	4	5°E	6°N
B	7	92	76	5.9	5	800	7	53	−27	6.2	7	144°E	0°N
C	5	39	42	3.5	3	–	–	–	–	–	–	–	–
D	4	–	–	1.6	–	200–400	4	10	11	3.4	5	184°E	28°N
E	4	239	41	3.6	8	300	4	200	17	3.9	42	355°E	−13°N
F	6	19	36	5.4	8	200–600	6	16	−26	5.5	10	180°E	8°N
G	4	–	–	1.9	–	400	4	163	−20	3.3	5	35°E	−32°N
						Mean of significant sites after reversing B, D and F:							
						Samples:	35	199	23	26.1	4	357°E	−10°N
									($d\psi$ = 8, dX= 15)				
						Sites:	6	195	15	5.2	6	360°E	−14°N
									($d\psi$ = 15, dX= 30)				

of $D = 195, I = 15°$ ($k = 6, \alpha_{95} = 28°$) giving a virtual geomagnetic pole at 0°E, 14°S.

4. Discussion

Unfortunately the low magnetic stability of the bulk of the collection from the Sulitjelma Gabbro has resulted in low within-site precisions, and the combined mean direction is not well defined. However the dual polarity and the distinction of the derived pole from any post-Carboniferous pole for Europe demonstrate that the magnetisation is at least Palaeozoic in origin and probably primary. Although local gabbro foliation in this area is variable and mostly steep, no dip correction has been applied to this result because there is abundant evidence (Dr. R. Mason, personal communication, 1973) that the gabbro cooled comparatively slowly, and was locally tectonised while still undergoing metamorphism to amphibolite facies.

Poorter's [4] preliminary Cambrian pole from the Fen Complex (600–530 m.y.) of southern Norway is at 63°N, 142°E and suggests an apparent polar shift of 55 degrees of arc between this time and the acquisition of the magnetisation of the Sulitjelma Gabbro which from the petrologic evidence appears to date from the regional metamorphism with a probable age of ca. 440 m.y. The path connecting these two poles is similar to that shown by Cambo-Ordovician results from the British Isles (Fig. 2), and indeed, the pole obtained from the Sulitjelma Gabbro is close to those obtained from contemporaneous material from Britain [17–20].

The broad agreement of the Sulitjelma pole with the British results suggests that Britain and Norway were part of the same crustal plate (Baltic/Russian plate of Briden et al. [3]) in Ordovician times or before the termination of the Caledonide folding. However, an unequivocal acceptance of this hypothesis is precluded by the major discrepancy between British and Norwegian Siluro-Devonian results. The behaviour of the palaeomagnetic pole for the British Isles in Late Silurian/Lower Devonian times is now well established [3]. The two well-defined Siluro-Devonian results from Norway are from the Roragen Complex of late Lower or Middle Devonian age [21] and the Kvamshesten formation [22] of Middle or early Upper Devonian age which give poles near British Lower Carboniferous results (Fig. 2). The discrepency between British and Norwegian Siluro-Devonian results may be only apparent, and due to inadequate age coverage. This is suggested by the general agreement of results from Middle-Upper Devonian lavas from the Orkney and Shetland Islands (Piper, in preperation) with the Norwegian results. With the agreement of the pole from the Sulitjelma Gabbro with poles of comparable age from the British Isles a plate

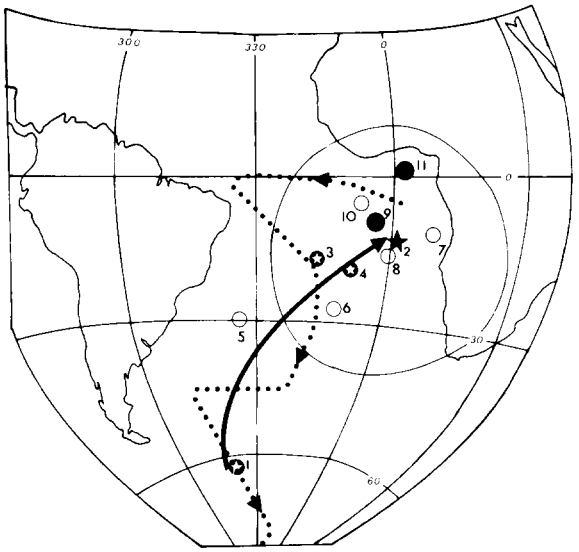


Fig. 2. Summary of Palaeozoic palaeomagnetic poles from Britain and Norway. Norwegian poles (stars) are: 1 = Fen Complex, Cambrian [4]; 2 = Sulitjelma Gabbro, Ordovician, this paper; 3 = Roragen Complex, Devonian [21]; 4 = Kvamshesten Formation, Devonian [22]. British Cambrian poles (circles) are: 5 = Canisp Porphyry, northwest Scotland, post-Lower Cambrian (C. Thomas and Piper, in preparation); 6 = Caerfai Series, Wales [24]. British Ordovician poles (circles) are: 7 = Younger Gabbros, Aberdeenshire [17]; 8 = Builth Volcanic Series [18]; 9 = Borrowdale Volcanic Series [19]; 10 = Treffgarne Volcanic Series [20]; and 11 = Builth Ashgillian intrusions [18]. The dashed path is the apparent polar wander path of Britain for post-Ordovician times [3].

tectonic interpretation [2] may not prove the correct explanation for discrepancies between Palaeozoic paleomagnetic results from Britain and Norway.

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